PROBLEM SET 7

1. Convert the following to minterms

   $ABC + AB + DC + \overline{D}$

   $ABCDE + AB\overline{E} + ACD$

$$
\begin{aligned}
ABC + AB + DC + \overline{D} &= 111X + 11XX + XX11 + XXX0 \\
&= 1110 + 1111 + 1100 + 1101 + 1110 + 1111 + 0011 + 0111 + 1011 \\
&\quad + 1111 + 0000 + 0010 + 0100 + 0110 + 1000 + 1010 + 1100 + 1110 \\
&= 1110 + 1111 + 1100 + 1101 + 0011 + 0111 + 1011 + 0000 + 0010 + \\
&\quad 0100 + 0110 + 1000 + 1010 \\
&= \Sigma m\ (0, 2, 3, 4, 6, 7, 8, 10, 11, 12, 13, 14, 15)
\end{aligned}
$$

$$
\begin{aligned}
ABCDE + AB\overline{E} + ACD &= 11111 + 11XX0 + 1X11X \\
&= 11111 + 11000 + 11010 + 11100 + 11110 + 10110 + 10111 + \\
&\quad 11110 + 11111 \\
&= ABCDE + AB\overline{C}\,\overline{D}\,\overline{E} + AB\overline{C}D\overline{E} + ABC\,\overline{D}\,\overline{E} + ABCD\overline{E} + A\overline{B}CD\overline{E} \\
&\quad + A\overline{B}CDE + ABCD\overline{E} + ABCDE \\
&= \Sigma m\ (31, 24, 26, 28, 30, 22, 23) \\
&= \Sigma m\ (22, 23, 24, 26, 28, 30, 31)
\end{aligned}
$$

2. Convert the following to maxterms
   a. $(A+B+\overline{D})\ (\overline{A}+C+D)\ (\overline{A}+\overline{D})$
   b. $A\ (\overline{A}+B)\ (\overline{C})$

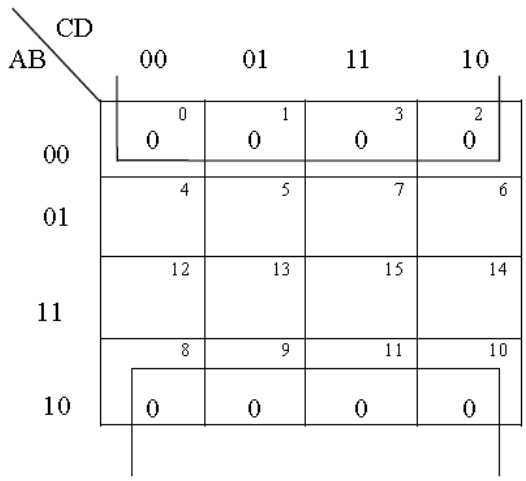
$$
\begin{aligned}
(A+B+\overline{D})\ (\overline{A}+C+D)\ (\overline{A}+\overline{D}) &= (00X1)\ (1X00)\ (1XX1) \\
&= (0001)\ (0011)\ (1000)\ (1100)\ (1001)\ (1011)\ (1101)\ (1111) \\
&= (0001)\ (0011)\ (1000)\ (1001)\ (1011)\ (1100)\ (1101)\ (1111) \\
&= \prod M\ (1, 3, 8, 9, 11, 12, 13, 15) \\
&= (A+B+C+\overline{D})\ (A+B+\overline{C}+\overline{D})\ (\overline{A}+B+C+D)\ (\overline{A}+B+C+\overline{D}) \\
&= (\overline{A}+B+\overline{C}+\overline{D})\ (\overline{A}+\overline{B}+C+D)\ (\overline{A}+\overline{B}+C+\overline{D})\ (\overline{A}+\overline{B}+\overline{C}+\overline{D})
\end{aligned}
$$

$$
\begin{aligned}
A\ (\overline{A}+B)\ (\overline{C}) &= (0XX)\ (10X)\ (XX1) \\
&= (000)\ (001)\ (010)\ (011)\ (100)\ (101)\ (001)\ (011)\ (101)\ (111) \\
&= (000)\ (001)\ (010)\ (011)\ (100)\ (101)\ (111) \\
&= \prod M\ (0, 1, 2, 3, 4, 5, 7) \\
&= (A+B+C)\ (A+B+\overline{C})\ (A+\overline{B}+C)\ (A+\overline{B}+\overline{C})\ (\overline{A}+B+C)\ (\overline{A}+B+\overline{C})\ (\overline{A}+\overline{B}+\overline{C})
\end{aligned}
$$

3. Reduce the following expressions using K-map

   $(\overline{A}+B)\ (A+B+\overline{D})\ (B+\overline{C})\ (B+C+D)$

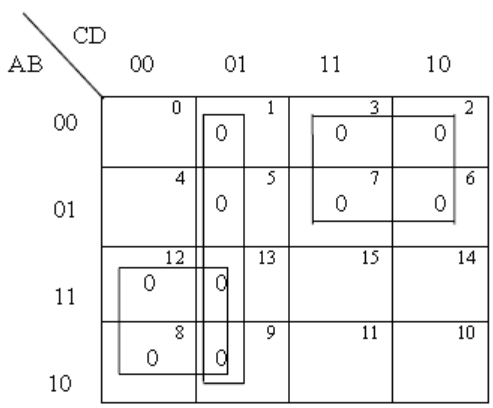
$$
\begin{aligned}
&(\overline{A}+B)\ (A+B+\overline{D})\ (B+\overline{C})\ (B+C+D) \\
&\quad = (10XX)\ (00X1)\ (X01X)\ (X000) \\
&\quad = (1000)\ (1001)\ (1010)\ (1011)\ (0001)\ (0011) \\
&\qquad (0010)\ (0011)\ (1010)\ (1011)\ (0000)\ (1000) \\
&\quad = \prod M\ (8, 9, 10, 11, 1, 3, 2, 3, 10, 11, 0, 8) \\
&\quad = \prod M\ (0, 1, 2, 3, 8, 9, 10, 11)
\end{aligned}
$$

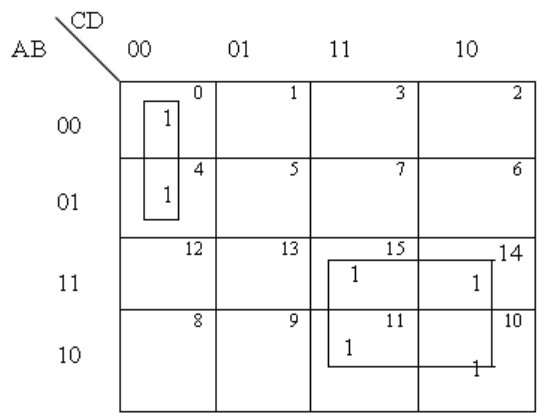


$f_{min} = B$

4. Reduce ∏M(1, 2, 3, 5, 6, 7, 8, 9, 12, 13) and implement it in universal logic.

$f = \prod M(1, 2, 3, 5, 6, 7, 8, 9, 12, 13) = \Sigma m\ (0, 4, 10, 11, 14, 15)$





$f_{min} = (\overline{A} + C)\ (C + \overline{D})\ (A + \overline{C})$ (9 gate i/ps)

$f_{min} = \overline{A}\ \overline{C}\ \overline{D} + AC$ (7 gate i/ps)

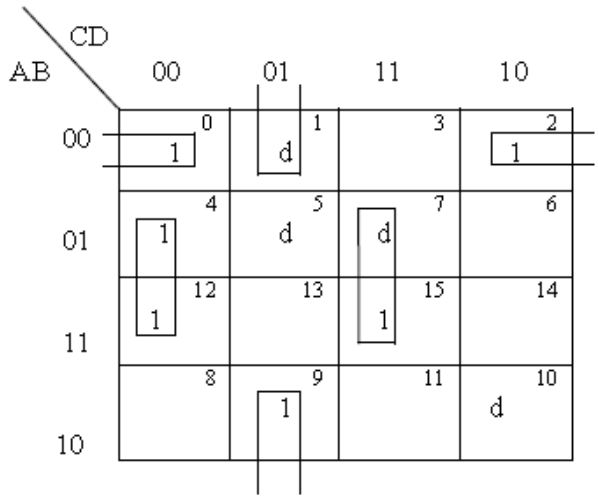
The SOP minimal is the real minimal because it requires only 7 gate inputs compared to 9 of POS minimal. The SOP minimal can be implemented using NAND logic.

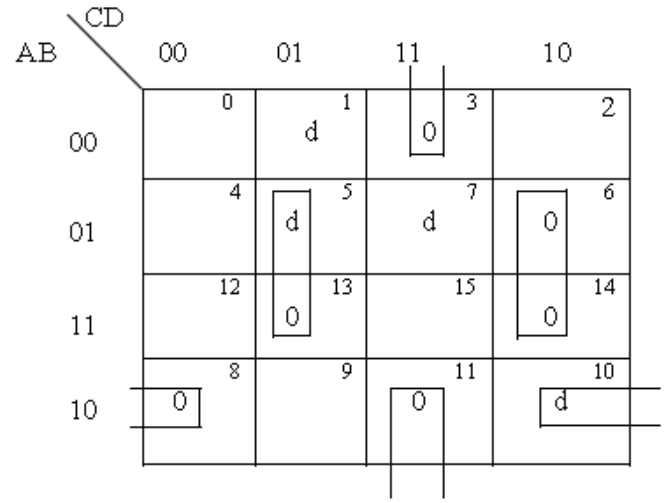
Therefore actual $f_{min} = \overline{A}\ \overline{C}\ \overline{D} + AC = \overline{\overline{A}\ \overline{C}\ \overline{D}}\ .\ \overline{AC}$

5. Reduce the following expressions using K-map and implement them in universal logic.
   (a) ∏M (3, 6, 8, 11, 13, 14) .d(1, 5, 7, 10)
   (b) Σm (0, 1, 4, 5, 6, 7, 9, 11, 15) + d (10, 14)

   (a)

(d) $f = \prod M(3, 6, 8, 11, 13, 14) . d(1, 5, 7, 10)$
$= \Sigma m(0, 2, 4, 9, 12, 15) + d(1, 5, 7, 10)$





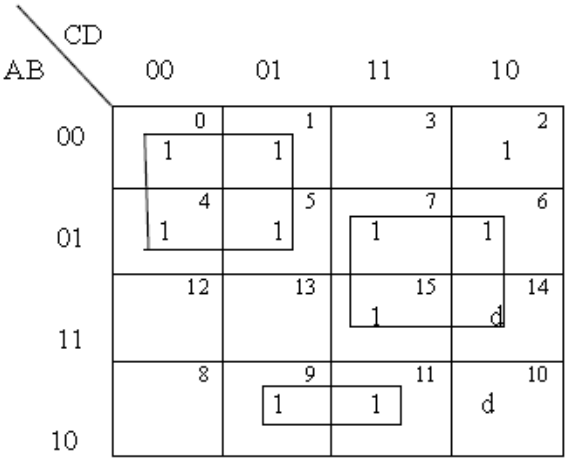
$f_{min} = \overline{B}\,\overline{C}\,D + BCD + B\,\overline{C}\,\overline{D} + \overline{A}\,\overline{B}\,\overline{D}$ (16 gate i/ps)
$f_{min} = (B + \overline{C} + \overline{D})\,(\overline{B} + C + \overline{D})\,(\overline{B} + \overline{C} + D)\,(\overline{A} + B + D)$ (16 gate i/ps)

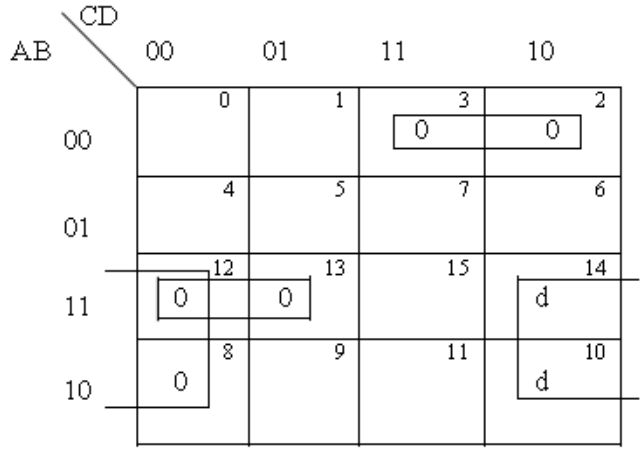
The SOP minimal and POS minimal require the same number of gate inputs. So any one form can be used. The SOP minimal can be implemented using NAND logic.

Actual $f_{min} = \overline{B}\,\overline{C}\,D + BCD + B\,\overline{C}\,\overline{D} + \overline{A}\,\overline{B}\,\overline{D} = \overline{\overline{(\overline{B}\,\overline{C}\,D)}\,.\,\overline{(BCD)}\,.\,\overline{(B\,\overline{C}\,\overline{D})}\,.\,\overline{(\overline{A}\,\overline{B}\,\overline{D})}}$

B)

$f = \Sigma m(0, 1, 4, 5, 6, 7, 9, 11, 15) + d(10, 14)$
$= \prod M(2, 3, 8, 12, 13) . d(10, 14)$





$f_{min} = \overline{A}\,\overline{C} + BC + A\overline{B}D$ (10gate i/ps)

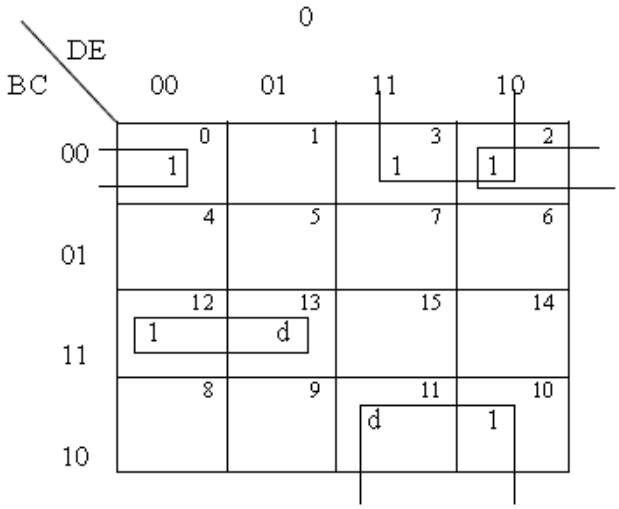
$f_{min} = (\overline{A} + D)(A + B + \overline{C})(\overline{A} + \overline{B} + C)$ (11 gate i/ps)

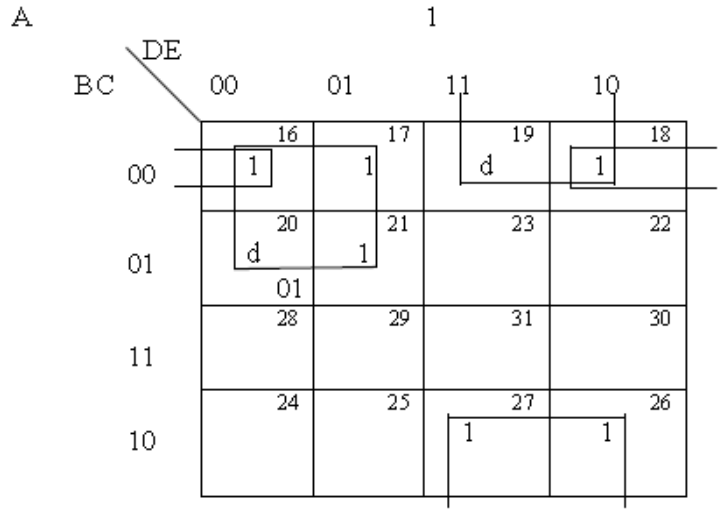
6. Simplify the following logic expressions and realize them using universal gates.
   a. Σm (0, 2, 3, 10, 12, 16, 17, 18, 21, 26, 27) + d (11, 13, 19, 20)
   b. Σm (4, 6, 8, 9, 10, 12, 13, 18, 19, 25, 26, 29, 33, 35, 36, 41, 42, 48, 49, 50, 56, 57) + d(0, 1, 11, 15, 30, 38, 40)

Solution:

(a)

SOP form f = Σm (0, 2, 3, 10, 12, 16, 17, 18, 21, 26, 27) + d (11, 13, 19, 20)

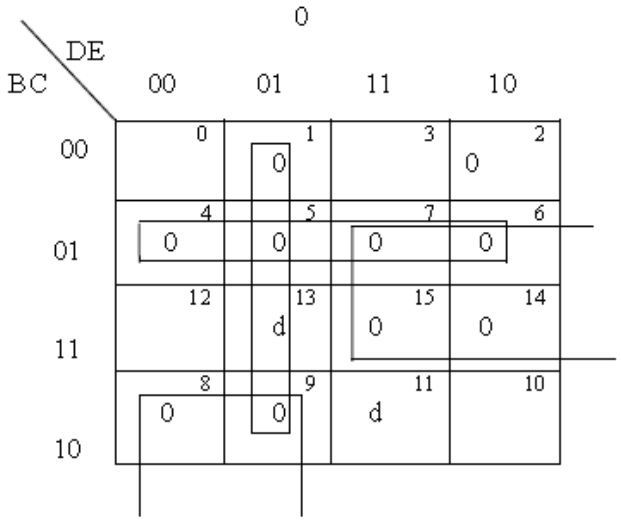


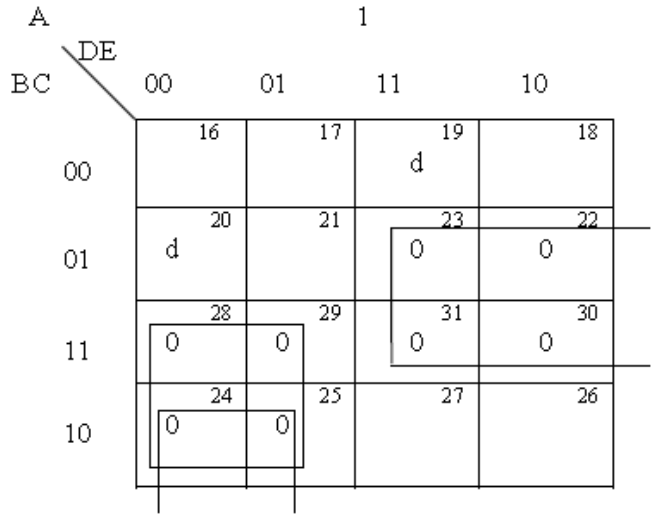


$f_{min} = \overline{C}D + A\overline{B}\,\overline{D} + \overline{B}\,\overline{C}\,\overline{E} + \overline{A}BC\overline{D}$ (16 gate inputs)

$= \overline{\overline{(\overline{C}D)}\,.\,\overline{(A\overline{B}\,\overline{D})}\,.\,\overline{(\overline{B}\,\overline{C}\,\overline{E})}\,.\,\overline{(\overline{A}BC\overline{D})}}$

POS form f = $\prod$M (1, 4, 5, 6, 7, 8, 9, 14, 15, 22, 23, 24, 25, 28, 29, 30, 31) . d(11, 13, 19, 20)





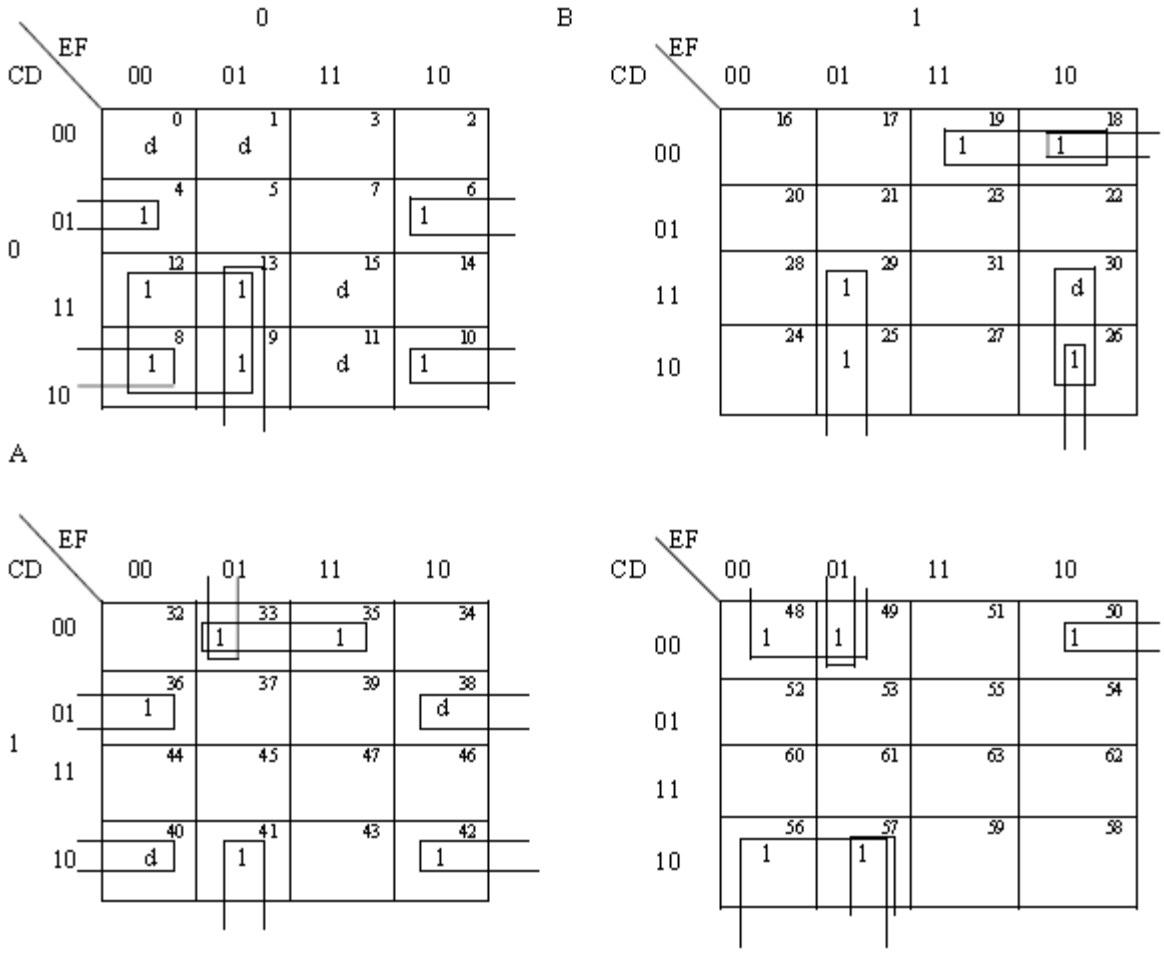
$$f_{min} = (\overline{C} + \overline{D})\,(A + D + \overline{E})\,(A + B + \overline{C})\,(\overline{B} + C + D)\,(\overline{A} + \overline{B} + D)\ \text{(19 gate inputs)}$$

$$= \overline{\overline{(\overline{C} + \overline{D})} + \overline{(A + D + \overline{E})} + \overline{(A + B + \overline{C})} + \overline{(\overline{B} + C + D)} + \overline{(\overline{A} + \overline{B} + D)}}$$

The SOP minimal is the real minimal because it requires only 16 gate inputs compared to 19 of POS minimal. The SOP minimal can be implemented using NAND logic.

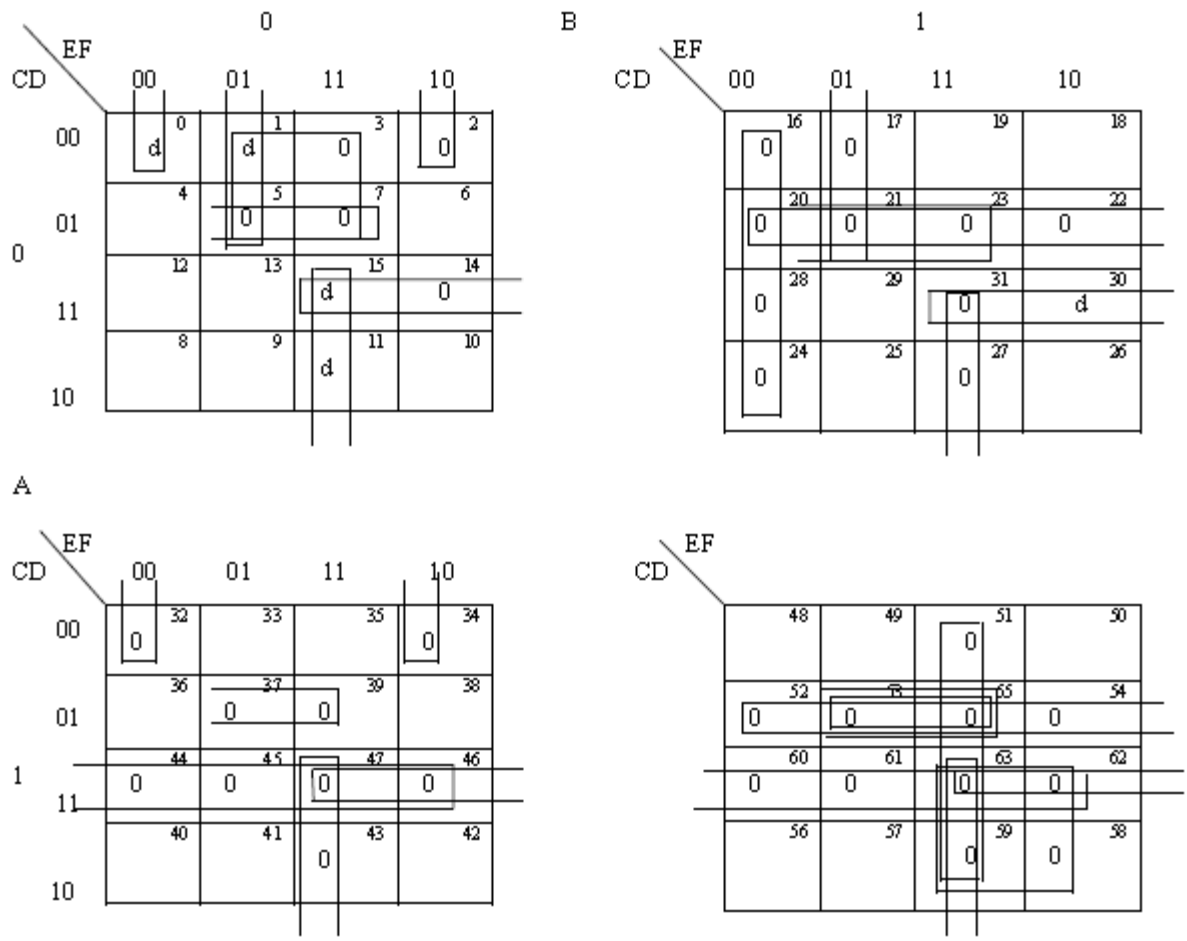
(b)

SOP form f = Σm (4, 6, 8, 9, 10, 12, 13, 18, 19, 25, 26, 29, 33, 35, 36, 41, 42, 48, 49, 50, 56, 57) + d(0, 1, 11, 15, 30, 38, 40)



$f_{min} = \overline{A}B\overline{C}\,\overline{D}E + \overline{B}\,\overline{C}D\overline{F} + \overline{B}C\overline{D}\,\overline{F} + AB\overline{D}\,\overline{E} + A\overline{B}\,\overline{C}\,\overline{D}F + \overline{A}C\overline{E}F + B\overline{C}\,\overline{D}E\overline{F} + \overline{A}\,\overline{B}C\overline{E}$
$+ \overline{A}BCE\overline{F} + A\overline{D}\,\overline{E}F$ (54 gate inputs).

POS form $f = \prod M(2, 3, 5, 7, 14, 16, 17, 20, 21, 22, 23, 24, 27, 28, 31, 32, 34, 37, 39, 43, 44, 45, 46, 47, 51, 52, 53, 54, 55, 58, 59, 60, 61, 62, 63)$. d(0, 1, 11, 15, 30, 38, 40)



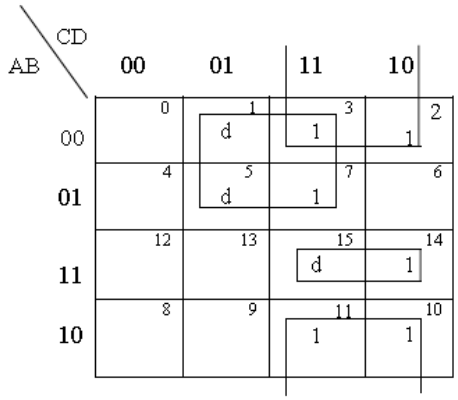
$f_{min} = (\overline{C} + \overline{E} + \overline{F})\,(\overline{C} + \overline{D} + \overline{E})\,(\overline{A} + \overline{B} + \overline{C} + \overline{E})\,(B + C + D + F)\,(\overline{B} + C + \overline{D})\,(\overline{A} + \overline{B} + \overline{C})$
$(A + B + \overline{E} + \overline{F})\,(C + \overline{D} + \overline{F})\,(A + \overline{B} + E + F)\,(A + B + C + \overline{F})\,(A + C + E + \overline{F})$ (50 gate inputs).

The POS minimal is the real minimal because it requires only 50 gate inputs compared to 54 of SOP minimal. The POS minimal can be implemented using NOR logic.

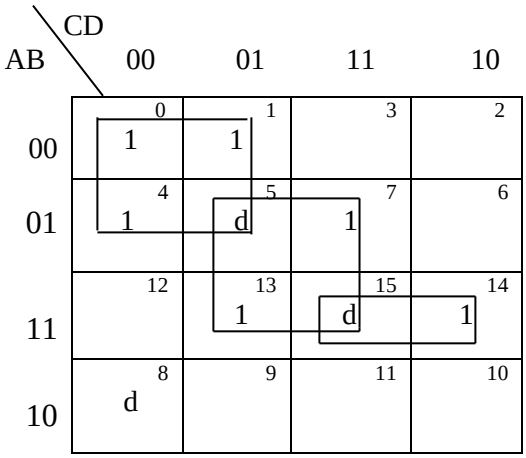
7. Minimize and implement the following multiple output functions in SOP form using K-maps.
   $f_1 = \Sigma$ m (2, 3, 7, 10, 11, 14) + d (1, 5, 15)
   $f_2 = \Sigma$ m (0, 1, 4, 7, 13, 14) + d (5,8,15)

   Hint: Use shared minterm and map to K-map
   Solution:
   $f_1 = \Sigma$ m (2, 3, 7, 10, 11, 14) + d (1, 5, 15)
   $f_2 = \Sigma$ m (0, 1, 4, 7, 13, 14) + d (5,8,15)

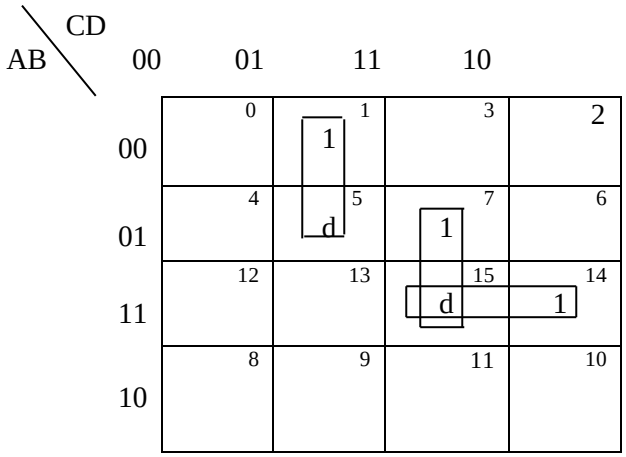
   $\mathbf{f_c = \Sigma\, m(1, 7, 14) + d\,(5, 15)}$



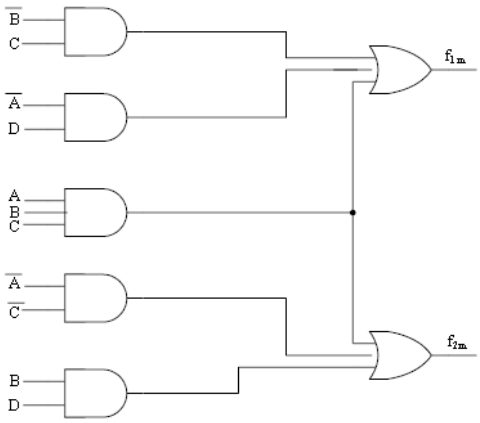
(a) $f_{1m} = \overline{B}\,C + A\,B\,C + \overline{A}\,D$



(b) $f_{2m} = \overline{A}\ \overline{C} + B\,D + A\,B\,C$



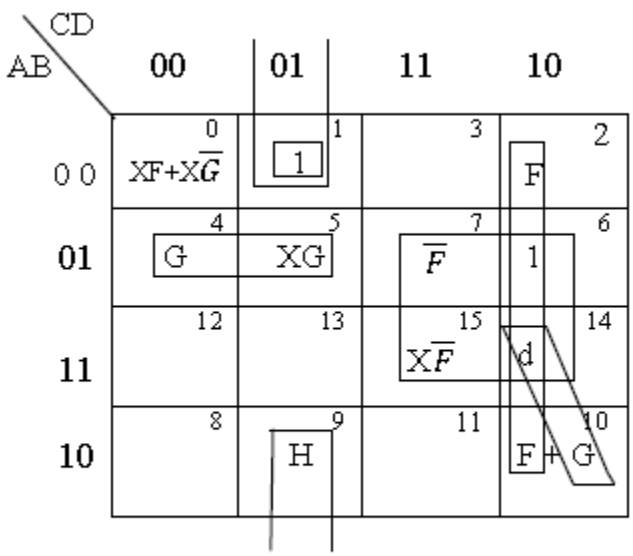
(c) $f_c = A\,B\,C + \overline{A}\ \overline{C}\,D + B\,C\,D$



8. Reduce the following expressions using a 4-variable K-map.
   a. S = m_1+ F m_2+ G m_4+ m_6+ $\overline{F}$ m_7+ H m_9 + (F + G) m_{10} +$d[(F+\bar{G})m_0 + Gm_5 + m_{14} + \bar{F}m_{15}]$
   b. T = E m_0+ H m_1+ F m_2+ ($\overline{E}$ + G) m_6 + m_8 + d (m_{10} + $\overline{E}$ m_{14} + G m_7)
   c. U = E m_3 + m_5 + m_7 + F m_{10} + m_{12} + m_{14}

Solution:

The given function is a 8-variable function. Its reduction using a 4-variable K-map is shown below.

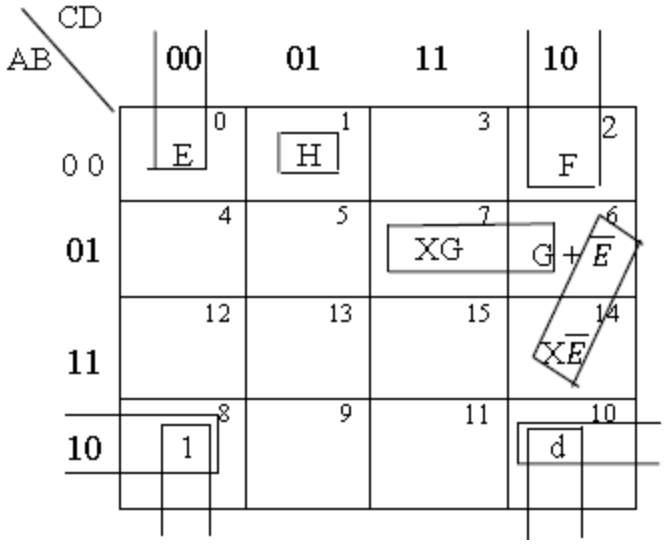


$S_{min} = \overline{B}\,\overline{C}\,D\,H + \overline{A}\,\overline{B}\,\overline{C}\,D + \overline{A}\,B\,\overline{C}\,G + C\,\overline{D}\,F + B\,C\,\overline{F} + A\,C\,\overline{D}\,G$

(b)

Given T = E m_0+ H m_1+ F m_2+ ($\overline{E}$ + G) m_6 + m_8 + d (m_{10} + $\overline{E}$ m_{14} + G m_7)

The given function is a 7-variable function. Its reduction using a 4-variable K-map is shown below.



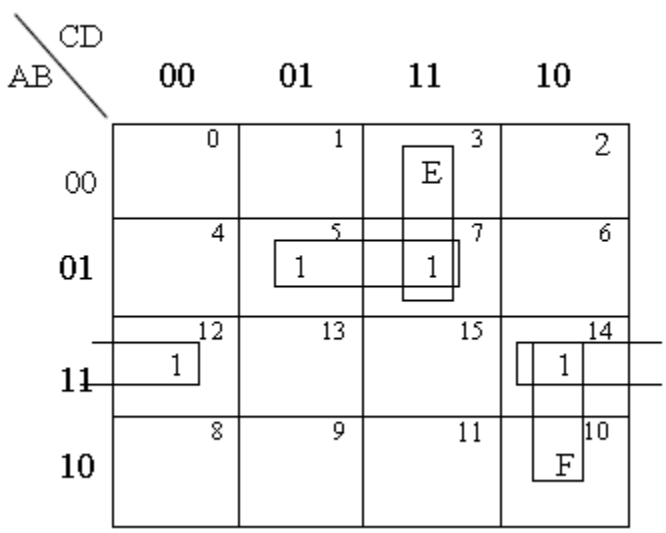
$T_{min} = \overline{B}\,\overline{C}\,\overline{D}\,E + \overline{B}\,C\,\overline{D}\,F + \overline{A}\,B\,C\,G + B\,C\,\overline{D}\,\overline{E} + A\,\overline{B}\,\overline{D} + \overline{A}\,\overline{B}\,\overline{C}\,D\,H$

Only E of m_8 is covered with E m_0. So m_8 and d m_{10} are combined.

(c)

Given $U = E\,m_3 + m_5 + m_7 + F\,m_{10} + m_{12} + m_{14}$

The given function is a 6-variable function. Its reduction using a 4-variable K-map is shown below.



$U_{min} = \overline{A}\,C\,D\,E + A\,C\,\overline{D}\,F + \overline{A}\,B\,D + A\,B\,\overline{D}$